SYNTOMIC MORPHISMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 5838-5841 and proof span 5843-5851 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof follows the author's reduction to the algebraic statement through the locality mechanism; no step was added.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The verification that being syntomic is a local property of ring maps, carried out in the same chapter's characterization of syntomic morphisms. Cited at proof lines 5843-5845. Allows the locality machinery for ring map properties to be applied to syntomicity.
- The transfer of a local property of ring maps to the corresponding property of morphisms, cited from the same chapter. Cited at proof lines 5845-5847. Turns the algebraic stability statement into the geometric one.
- Syntomicity of a ring map is stable under base change, cited from the algebra chapter. Cited at proof lines 5847-5849. Supplies the algebraic input of the proof.

Source reference. Original lemma 01UI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2411–2411. Same-chapter reference used in the retained author text at source line(s) 5813, 5850. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2352–2352. Same-chapter reference used in the retained author text at source line(s) 2720, 3908, 4144, 4169, 5832, 5847, 7112, 7127,

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7711, 8129, 8144. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2406–2406. Same-chapter reference used in the retained author text at source line(s) 5829, 5844. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *The base change of a morphism which is syntomic is syntomic.*

Proof. In the proof of Lemma 01UD we saw that being syntomic is a local property of ring maps. Hence the lemma follows from Lemma 01SV combined with the fact that being syntomic is a property of ring maps that is stable under base change, see Algebra, Lemma 01UI. □